

Ordinal and Latent-Class Modelling of Radiologist-Assessed Lung Nodule Malignancy

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Abstract

This project models the malignancy rating that radiologists assign to lung nodules on CT scans, using the LIDC-IDRI database. Each nodule is rated on an ordered 1–5 scale by up to four radiologists, who frequently disagree. We extract the radiologists' annotations with the `pylidc` library, derive geometric shape features from the drawn contours, and aggregate the 6859 annotations into 2637 nodules from 870 patients.

Exploratory analysis of the training set shows that nodule size separates the upper malignancy grades sharply but barely distinguishes grades 1 to 3, that the size measures are almost perfectly collinear, and that agreement between radiologists is only fair (Fleiss' $\kappa = 0.21$). These findings motivate the planned analysis: false-discovery-rate screening of features, variable selection, a proportional odds model, and a latent-class extension fitted by the EM algorithm in which the true malignancy grade is unobserved.

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1 Introduction

Lung cancer is commonly first detected as a small nodule on a computed tomography (CT) scan. Most nodules are benign, so radiologists grade how suspicious each one looks, and that grade determines follow-up. In a public health system with limited specialist time, a statistical model that predicts this grade from measurable properties of the nodule could help prioritise scans for review.

The Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI)[1] provides such grades. Up to four radiologists independently outlined every nodule of at least 3 mm and rated its likelihood of malignancy on an ordered scale from 1 (highly unlikely) to 5 (highly suspicious). The response is therefore ordinal, with categories that are ranked but not equally spaced, and it is observed several times per nodule with considerable disagreement.

We address three research questions.

1. Which geometric properties of a nodule are associated with its malignancy rating, once the false discovery rate is controlled across the candidate features?
2. Does a proportional odds model[10] adequately describe that association?
3. Does explicitly modelling disagreement between radiologists, by treating the true grade as a latent variable estimated with the EM algorithm[6, 5], change the estimated effects?

This document records the work completed so far: data acquisition, cleaning, validation and exploratory analysis. Section 5 outlines the modelling that follows.

2 Data Description

2.1 Source and Structure

LIDC-IDRI is hosted by The Cancer Imaging Archive[4]. The full image collection is about 133 GB, but the radiologists' annotations are distributed as XML and are bundled, already parsed, with the Python library `pylidc`[8]. Every quantity used in this project is computed from these annotations, so no CT images were downloaded.

The data are hierarchical:

$$\text{patient} \rightarrow \text{scan} \rightarrow \text{nodule} \rightarrow \text{annotation}.$$

The database contains 1018 scans, of which 883 (from 875 patients) contain at least one nodule annotation; 8 patients have two scans. An *annotation* is one radiologist's outline and ratings of one nodule. Annotations of the same physical nodule are grouped with `pylidc`'s clustering routine, giving 2651 nodules with 1 to 4 annotations each. The nodule, not the annotation, is our unit of analysis.

2.2 Variables

For each annotation we record the following.

- **Malignancy** (response): ordinal rating 1–5.
- **Semantic ratings**: subtlety, sphericity, margin, lobulation, spiculation and texture, each an ordinal 1–5 rating by the same radiologist. Calcification (1–6) and internal structure (1–4) are nominal categories.
- **Geometric features from `pylidc`**: diameter (mm), surface area (mm^2) and volume (mm^3) of the outlined region.
- **Shape features from the binary mask**: height of the bounding box (mm), number of slices spanned, elongation and flatness (square roots of the ratios of the second and third eigenvalues of the voxel-coordinate covariance to the first), convex-hull volume, and solidity (voxels in the nodule divided by voxels in its convex hull).

Intensity-based texture features (for example, grey-level co-occurrence statistics) require the CT images and are therefore not available. The candidate predictors are the geometric and shape features, and, as a secondary analysis, the semantic ratings.

3 Data Processing

3.1 Cleaning and Aggregation

Over-merged clusters. LIDC-IDRI used four readers per scan, so a cluster of more than four annotations means that neighbouring lesions were merged by the clustering routine. The 14 such clusters (81 annotations) were removed, leaving 2637 nodules from 870 patients.

Aggregation to nodules. For each nodule, geometric and shape features are averaged over readers, since each reader drew a slightly different outline. Ordinal semantic ratings are summarised by their median. Every reader's malignancy rating is kept, both as

columns $R_{i1}, \ldots, R_{i4}$ and as counts

$$C_{ir} = \#\{\, j : R_{ij} = r \,\}, \qquad r = 1, \ldots, 5,$$

which are the input to the latent-class model.

Nominal ratings. Calcification and internal structure are category labels, not rankings, and are almost constant: 6819 of the 6859 annotations record soft-tissue internal structure. Both are excluded from modelling and retained only in the master table. One annotation records an internal-structure value of 5, outside the documented range; it has no effect on the analysis.

Median ties. A single response per nodule is needed for screening and for the baseline model, and we use the median rating. When an even number of readers disagree on the middle two ratings, the median falls between grades; for example, ratings $(2, 3, 4, 4)$ give 3.5. Table 1 shows that this occurs for 548 nodules (20.8%). We round such medians *up*, which is the conservative choice for triage, and flag them. The latent-class model uses all ratings and needs no such rule.

Readers	1.5	2.5	3.5	4.5	Total
2	30	141	39	6	216
4	6	134	128	64	332

Table 1: Nodules whose median malignancy lies between two grades

Transformations. Volume, surface area and convex-hull volume are strongly right-skewed and are log-transformed. We also define compactness $= V^{2/3}/S$ for volume V and surface area S. Predictors are stored unstandardised; they are standardised within each modelling step using training-set means and standard deviations only.

3.2 Train/Test Split

Nodules of the same patient are not independent, so the data are split at the patient level: 70% of patients (1809 nodules) form the training set and 30% (828 nodules) the test set. All exploratory analysis, screening and model selection use the training set; the test set is reserved for the final evaluation.

3.3 Validation

An automated check script verifies the extracted and processed tables. It confirms, among other things, that every rating lies in its documented range, that no values are infinite, that the counts C_{ir} match the individual ratings, that the median and tie flags follow the rule above, that the aggregated features equal the reader means and medians recomputed from the raw file, and that no patient appears in both the training and test sets. All checks pass. Re-running the cleaning step reproduces the processed files exactly.

4 Exploratory Analysis

All results in this section use the 1809 training nodules.

4.1 Distribution of the Response

Figure 1 shows that grade 3 ("indeterminate") dominates: it is the median grade of 45.5% of nodules and accounts for 37.5% of all individual ratings. The extreme grades are used more often by individual readers than by the median, which is expected when readers disagree.

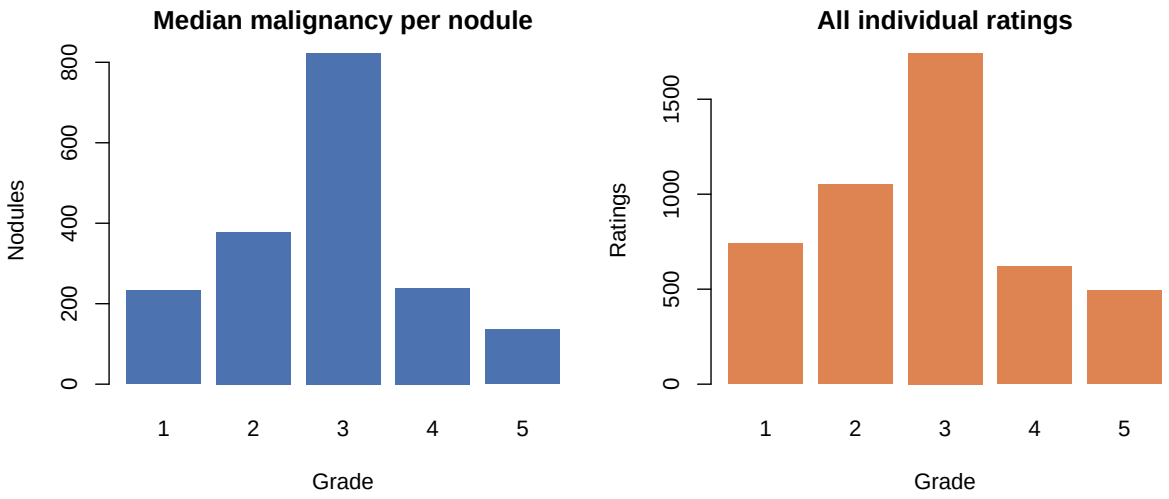


Figure 1: Median malignancy per nodule (left) and all individual ratings (right), training set

4.2 Features by Malignancy Grade

Table 2 summarises the geometric features and Fig. 2 shows them by median grade. Size is the dominant signal, but its effect is far from linear: the median diameter is 6.4, 6.7

and 7.4 mm for grades 1, 2 and 3, then rises to 13.9 and 22.8 mm for grades 4 and 5. Grade 1 nodules are even slightly larger than grade 2 nodules; a plausible explanation is that calcified nodules, which are benign regardless of size, are rated 1. Solidity and compactness decrease with grade, so suspicious nodules are less convex and less sphere-like, while elongation and flatness show no clear trend.

Feature	**Mean**	**Std. Dev.**	**Minimum**	**Maximum**
Diameter (mm)	10.137	6.818	2.586	49.945
log Volume	5.118	1.394	2.442	10.345
log Surface area	5.181	1.077	2.799	9.108
log Convex volume	5.111	1.553	1.653	10.812
Height (mm)	7.564	5.782	1.000	57.500
Elongation	0.753	0.121	0.306	0.988
Flatness	0.555	0.164	0.000	0.921
Solidity	0.795	0.106	0.228	1.000
Compactness	0.177	0.044	0.058	0.312

Table 2: Summary statistics of the geometric features, training set

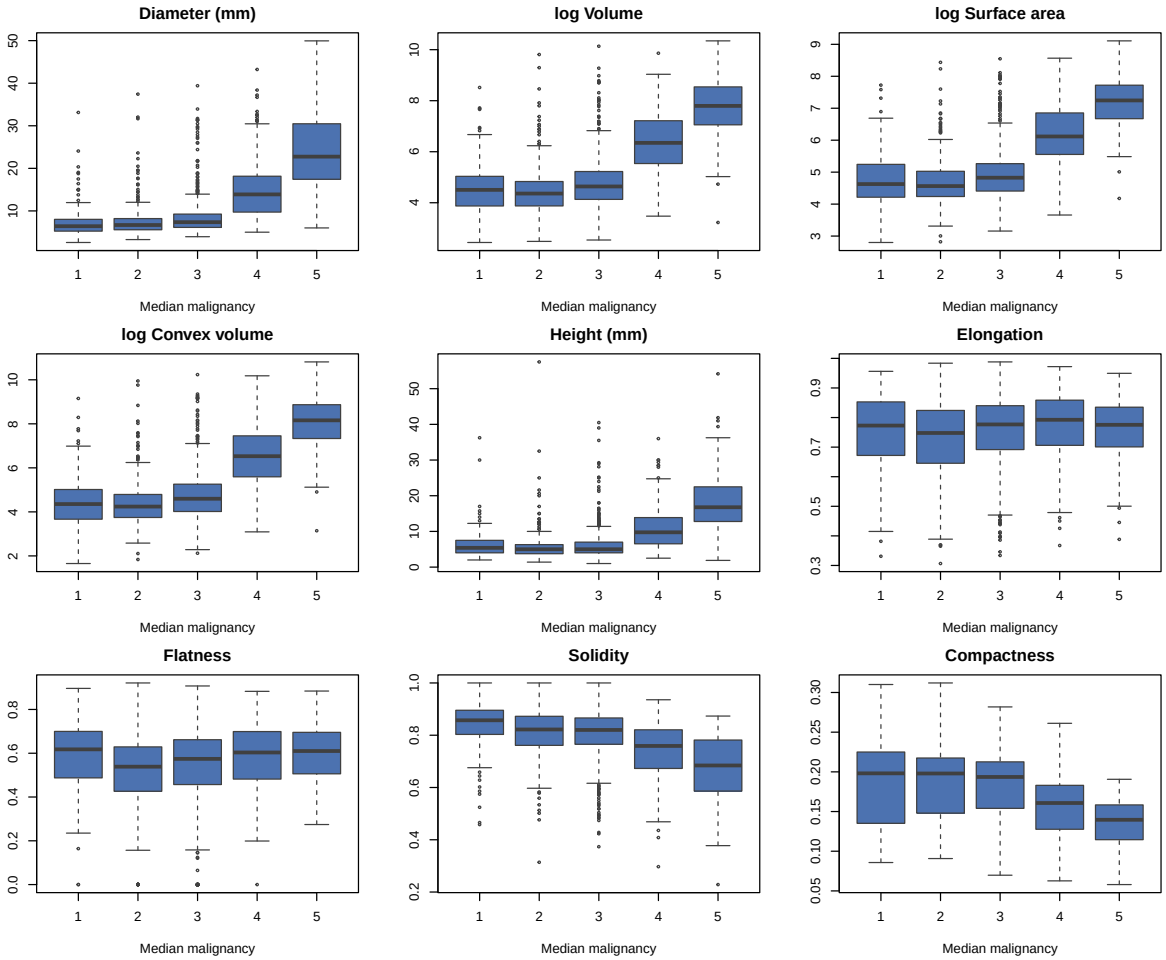


Figure 2: Geometric features by median malignancy grade, training set

4.3 Collinearity

Figure 3 shows Spearman correlations between the features. Diameter, log volume, log surface area and log convex volume are correlated at 0.91 to 1.00; they measure the same thing, and at most one should enter the model. Elongation and flatness are nearly uncorrelated with size and carry separate information.

The number of slices a nodule spans depends on the scan protocol as well as on the nodule: its correlation with slice spacing is -0.52, because thinner slices give more slices for the same nodule. It is therefore excluded in favour of the physical height in millimetres.

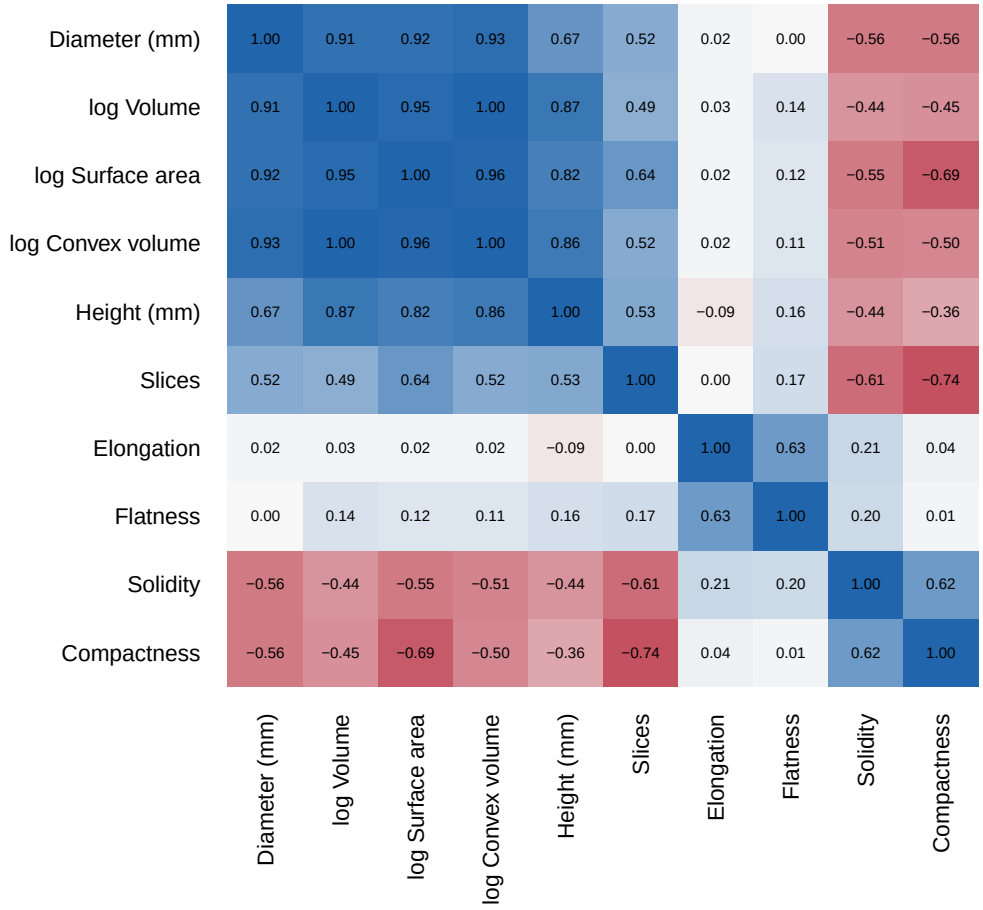


Figure 3: Spearman correlation between geometric features, training set

4.4 Agreement Between Radiologists

Of the 1285 training nodules rated by two or more radiologists, only 288 (22.4%) received identical ratings, and 151 have ratings three or more grades apart (Table 3 and Fig. 4).

Readers	**0**	**1**	**2**	**3**	**4**	**Total**
2	147	137	50	5	0	339
3	63	160	93	19	4	339
4	78	176	230	104	19	607
Total	288	473	373	128	23	1285

Table 3: Range (maximum minus minimum) of the malignancy ratings of each multi-reader nodule, training set

To summarise agreement we use Fleiss' κ[7], which allows the number of raters n_i to vary between nodules. The proportion of agreeing rater pairs for nodule i, the overall rating

proportions, and κ are

$$P_i = \frac{\sum_r C_{ir}(C_{ir}-1)}{n_i(n_i-1)}, \qquad p_r = \frac{\sum_i C_{ir}}{\sum_i n_i}, \qquad \kappa = \frac{\bar{P}-P_e}{1-P_e}, \quad P_e = \sum_r p_r^2,$$

where $\bar{P}$ is the mean of P_i over multi-reader nodules. We obtain $\bar{P} = 0.400$, $P_e = 0.240$ and $\kappa = 0.210$, which is only "fair" agreement on the scale of Landis and Koch[9].

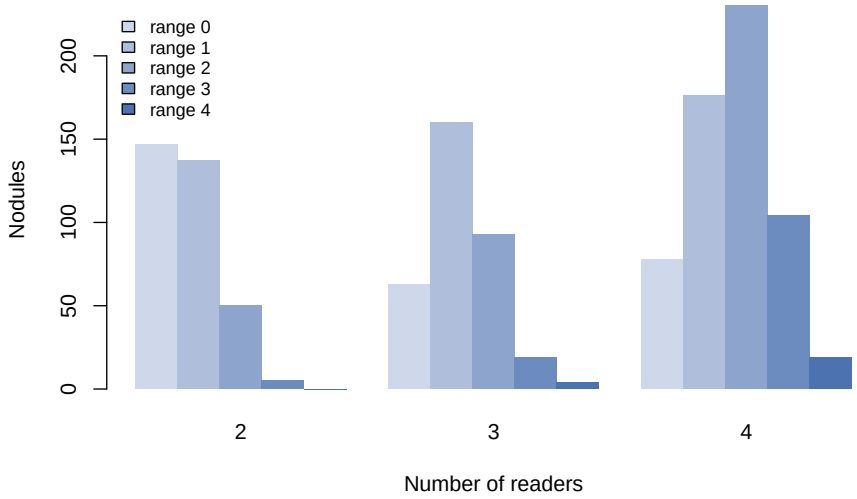


Figure 4: Range of malignancy ratings by number of readers, training set

4.5 Semantic Ratings

The Spearman correlation of the median malignancy with spiculation (0.42) and lobulation (0.40) is almost as large as with diameter (0.51). These ratings, however, come from the same radiologists who rate malignancy, so a model using them partly predicts a radiologist's opinion from the same radiologist's other opinions. They are therefore reserved for a secondary model.

5 Implications and Planned Analysis

The exploratory analysis shapes the remaining work as follows.

1. **Screening.** Each candidate feature will be tested for association with malignancy using a univariate proportional odds likelihood ratio test, and the Benjamini–Hochberg procedure[2] will control the false discovery rate. Because the features are strongly correlated, the Benjamini–Yekutieli correction will be reported as well.
2. **Selection.** Only one size variable can be retained (Fig. 3). AIC, BIC and the ordinal Lasso will be compared to fix the final design matrix.

3. **Proportional odds model.** With $K = 5$ grades,

$$\operatorname{logit} P(Y_i \le k \mid x_i) = \alpha_k - x_i^{\mathsf{T}}\beta, \qquad k = 1, \dots, 4,$$

so e^{β_j} is the common odds ratio of a higher grade per unit increase in x_j. The non-linear size pattern in Fig. 2 suggests that the proportional odds assumption may fail, and it will be tested with the Brant test[3] and a likelihood ratio test against a partial proportional odds model.

4. **Latent-class model.** The low agreement ($\kappa = 0.21$) motivates treating the true grade $Z_i \in \{1, \dots, 5\}$ as unobserved, with the proportional odds model for Z_i and a shared confusion matrix $\pi_{kr} = P(R_{ij} = r \mid Z_i = k)$ for the readers, in the spirit of Dawid and Skene[5]. The model will be fitted by the EM algorithm and compared with the median-rating model, both through likelihood ratio tests and on the test set.

A Appendix

A.1 Dataset

The LIDC-IDRI database[1] is available from The Cancer Imaging Archive[4]:

- TCIA: cancerimagingarchive.net/collection/lidc-idri

The annotations were read with `pylidc` version 0.2.3[8].

A.2 Code

The analysis is organised as a pipeline of scripts, run from the repository root.

- `python/extract.py`: reads the annotations with `pylidc` and computes the shape features (`data/raw/annotations.csv`).
- `R/01_clean.R`: aggregates to nodules, applies the rules of Section 3 and splits the data (`data/processed/`).
- `R/check_data.R`: validation checks.
- `R/eda.R`: exploratory analysis and figures.

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